INTRODUCTION TO INFORMATION GEOMETRY

Relative Entropy and Prediction

Week 6 · Lecture 1

Christopher Lee

Predicting observations with a model

Suppose observations follow the law $Q = (q_i)$, while our predictive model is $P = (p_i)$. Under logarithmic loss, outcome i costs $-\ln p_i$. Its expected loss is the cross-entropy

$$\mathrm{CE}(Q, P) = -\sum_i q_i \ln p_i.$$

Relative entropy is excess log loss

Using the generating law itself gives expected loss $H(Q)$. The additional loss from using P is

$$D(Q\|P) = \mathrm{CE}(Q, P) - H(Q)$$

$$= \sum_i q_i \ln \frac{q_i}{p_i}.$$

The first argument supplies the averaging probabilities.

The cost of an equal-probability prediction

Take the generating law $Q = (\frac{3}{4}, \frac{1}{4})$ and model $P = (\frac{1}{2}, \frac{1}{2})$. Compute

$$D(Q\|P) = \frac{3}{4} \ln \frac{3/4}{1/2} + \frac{1}{4} \ln \frac{1/4}{1/2}.$$

The cost of an equal-probability prediction

Take the generating law $Q = (\frac{3}{4}, \frac{1}{4})$ and model $P = (\frac{1}{2}, \frac{1}{2})$. The relative entropy is

$$D(Q\|P) = \frac{3}{4}\ln\frac{3}{2} + \frac{1}{4}\ln\frac{1}{2} \approx 0.1308.$$

This is the excess expected loss in nats per observation.

The direction of the comparison matters

Reversing the two arguments changes the averaging law. For $P = (\frac{1}{2}, \frac{1}{2})$, $Q = (\frac{3}{4}, \frac{1}{4})$,

$$D(P\|Q) = \frac{1}{2}\ln\frac{2}{3} + \frac{1}{2}\ln 2 = \frac{1}{2}\ln\frac{4}{3}.$$

$$D(P\|Q) \approx 0.1438 \neq D(Q\|P) \approx 0.1308.$$

KL divergence is a directed comparison.

Why divergence is nonnegative

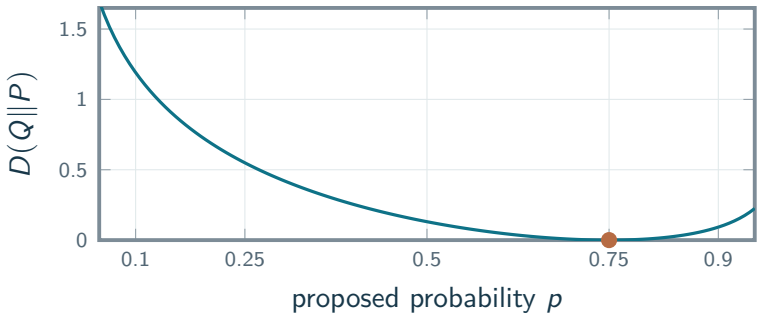
The inequality $\ln u \le u - 1$ holds for every $u > 0$. Apply it to $u = p_i / q_i$, multiply by q_i, and sum:

$$\begin{aligned} D(Q\|P) &\ge \sum_i q_i \left(1 - \frac{p_i}{q_i}\right) \\ &= \sum_i (q_i - p_i) = 0. \end{aligned}$$

Equality holds precisely when $P = Q$.

Divergence on the Bernoulli interval

Fix $Q = (.75, .25)$; vary $P = (p, 1-p)$.



Support determines finiteness

If the generating law assigns positive probability to an outcome that the predictive law assigns probability zero, then

$$D(Q\|P) = +\infty.$$

Finite divergence requires

$$\operatorname{supp}(Q) \subseteq \operatorname{supp}(P).$$

A boundary comparison

Let $Q = (1, 0)$ and $P = (\frac{1}{2}, \frac{1}{2})$. Calculate $D(Q\|P)$ and determine $D(P\|Q)$.

$$D(Q\|P) = ?, \qquad D(P\|Q) = ?.$$

A boundary comparison

Let $Q = (1, 0)$ and $P = (\frac{1}{2}, \frac{1}{2})$. The directed comparisons are

$$D(Q\|P) = \ln 2, \qquad D(P\|Q) = +\infty.$$

The second model rules out a possible generated outcome.

Observed frequencies become a model

If a sample contains counts N_i with total N, let $\widehat{q}_i = N_i/N$. The likelihood of the observed ordered sequence is

$$L(P) = \prod_i p_i^{N_i}.$$

Its average negative log likelihood is

$$-\frac{1}{N} \ln L(P) = H(\widehat{Q}) + D(\widehat{Q} \| P).$$

A Bernoulli likelihood

In 20 independent Bernoulli trials, 8 successes occur. The log likelihood for success probability p is

$$\ell(p) = 8 \ln p + 12 \ln(1 - p).$$

Find the maximum-likelihood estimate $\widehat{p}$.

A Bernoulli likelihood

In 20 independent Bernoulli trials, 8 successes occur. The derivative is

$$\ell'(p) = \frac{8}{p} - \frac{12}{1-p}.$$

Setting it to zero gives $\widehat{p} = 8/20 = 0.4$.

A hint of local geometry

Near a model fitted to the data, relative entropy is locally curved. For a Bernoulli law with parameter q, a small change to $q + u$ has a cost proportional to u^2. This second-order behavior will supply a metric for probability laws.